

Nonconventional Energy sources (BT EE 707)

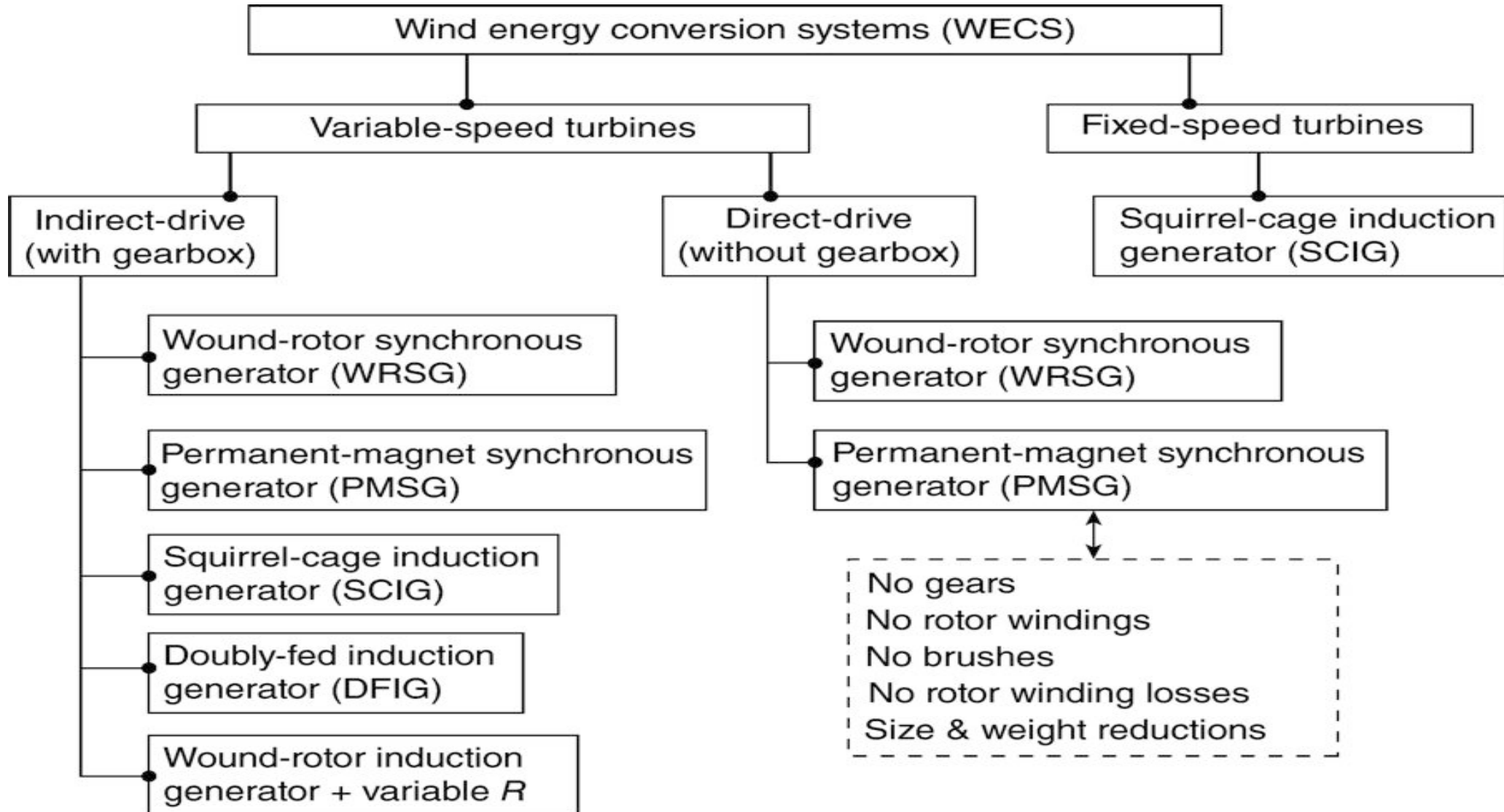
Wind Power



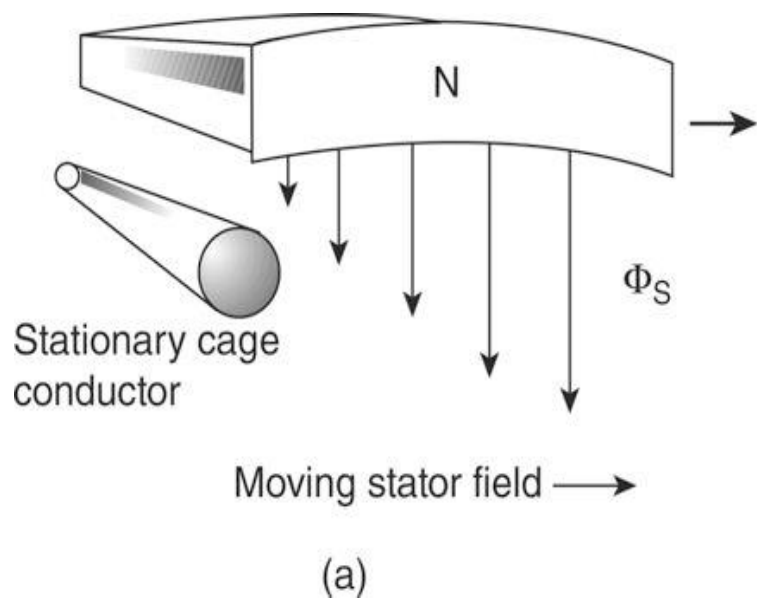
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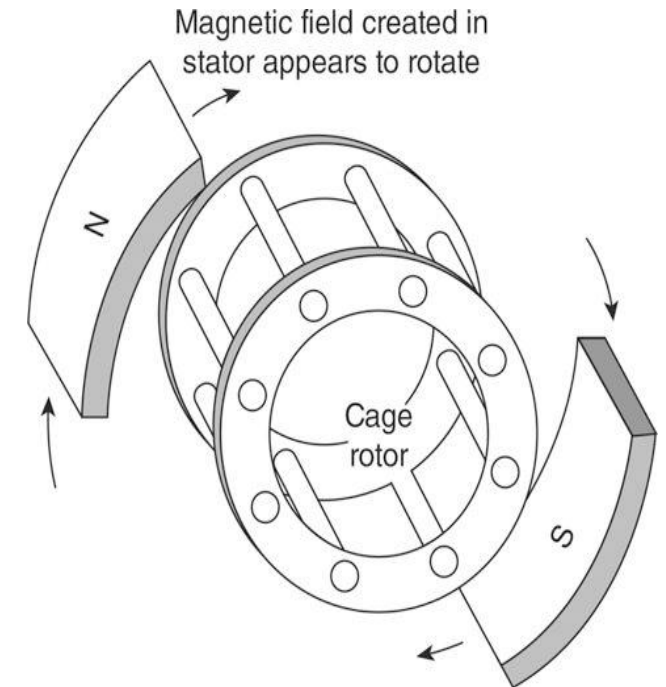
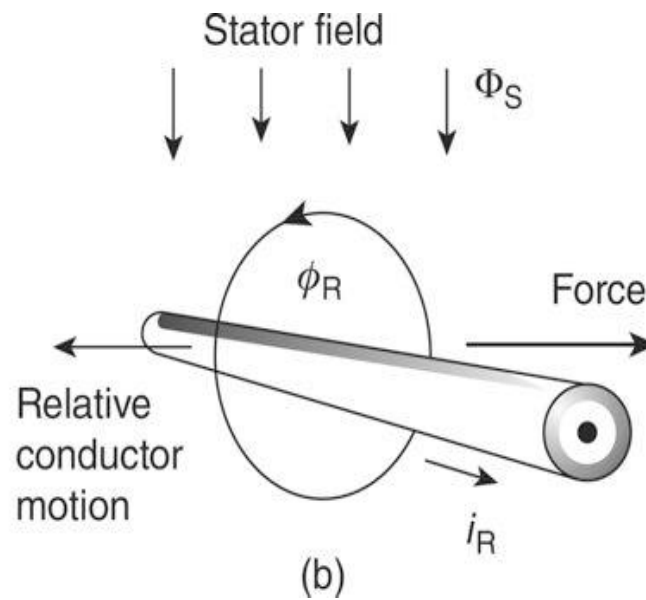
Wind Energy Conversion Systems



Squirrel-Cage Induction Generator

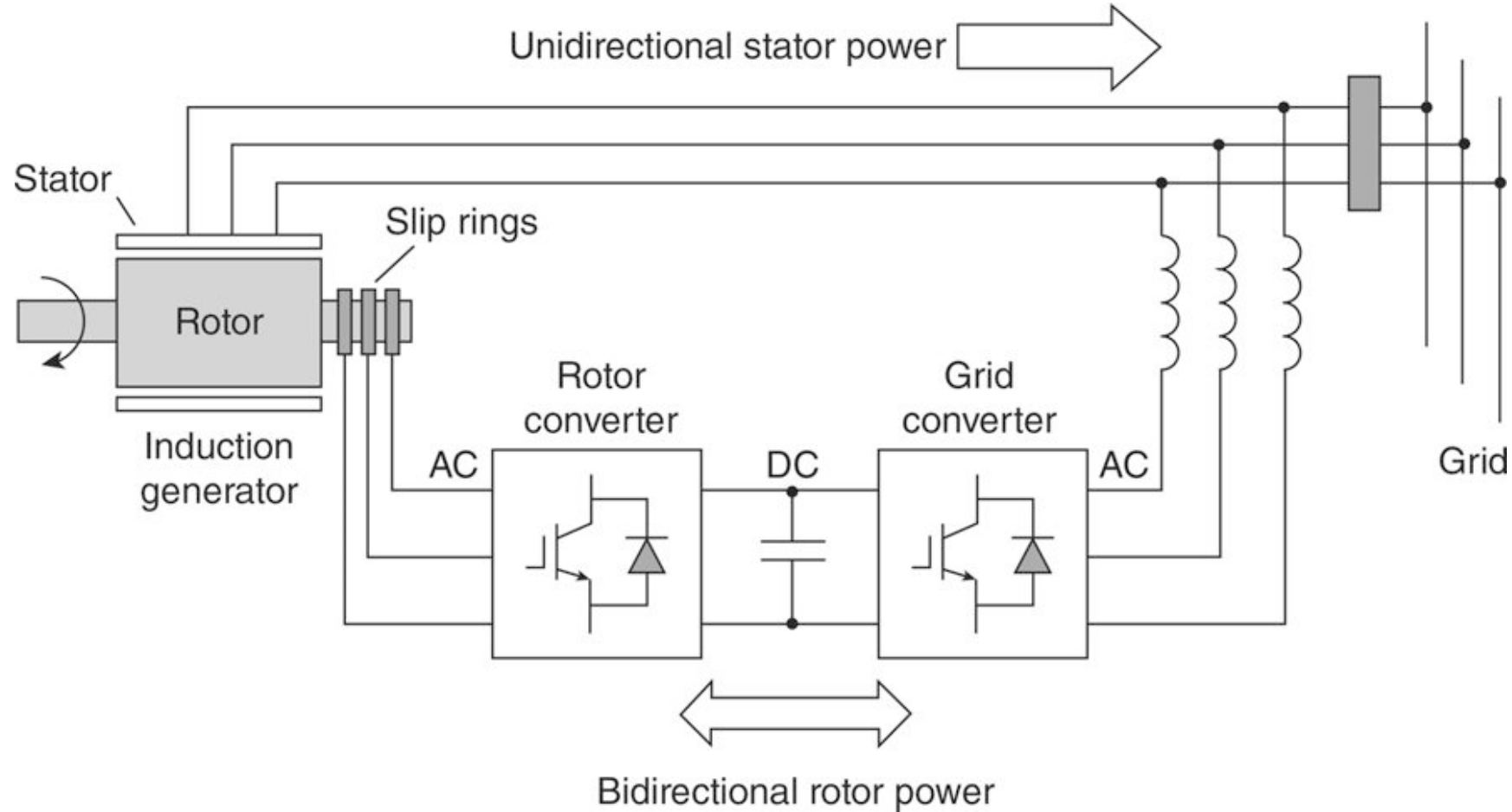


Cage Conductor Force & Current



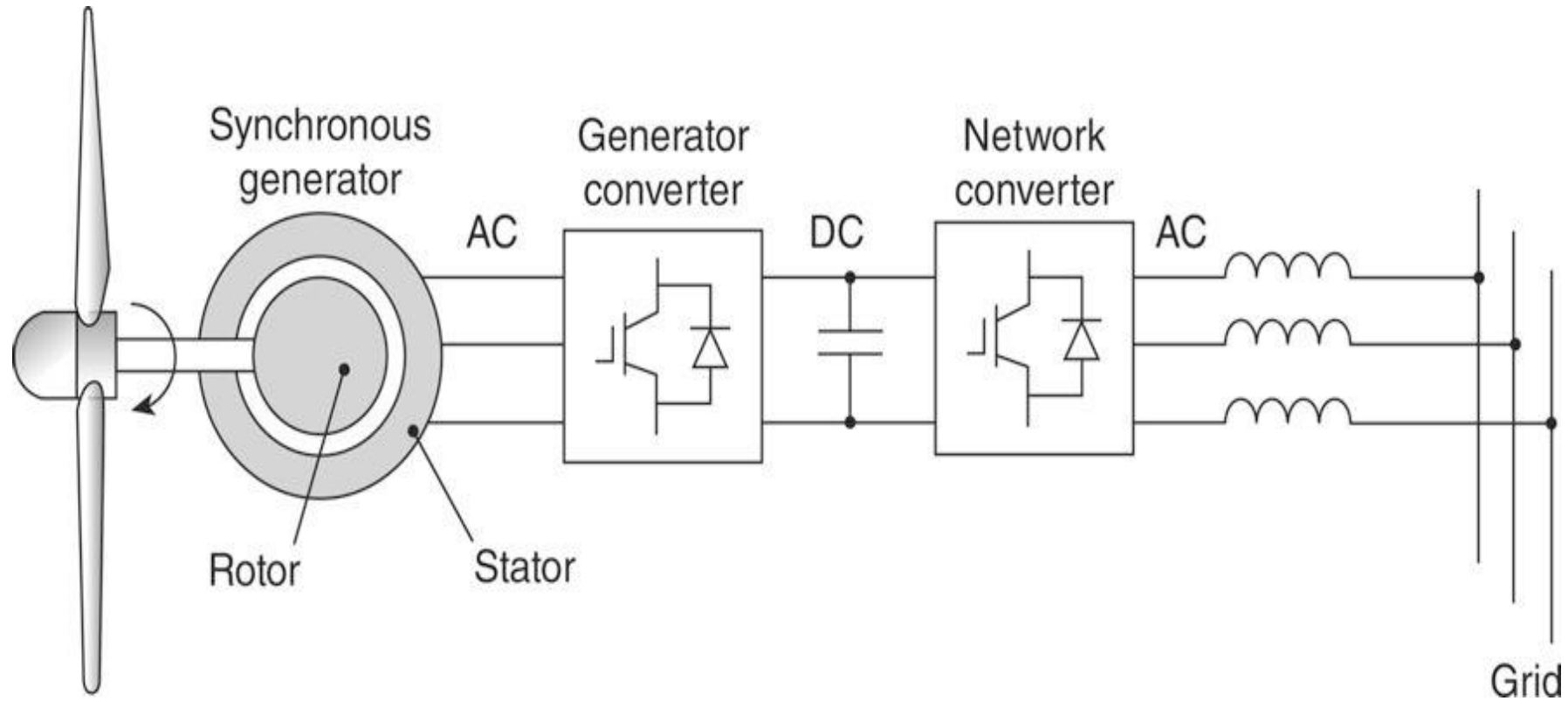
Squirrel-Cage Induction Generator

Doubly-fed Induction Generator (DFIG)



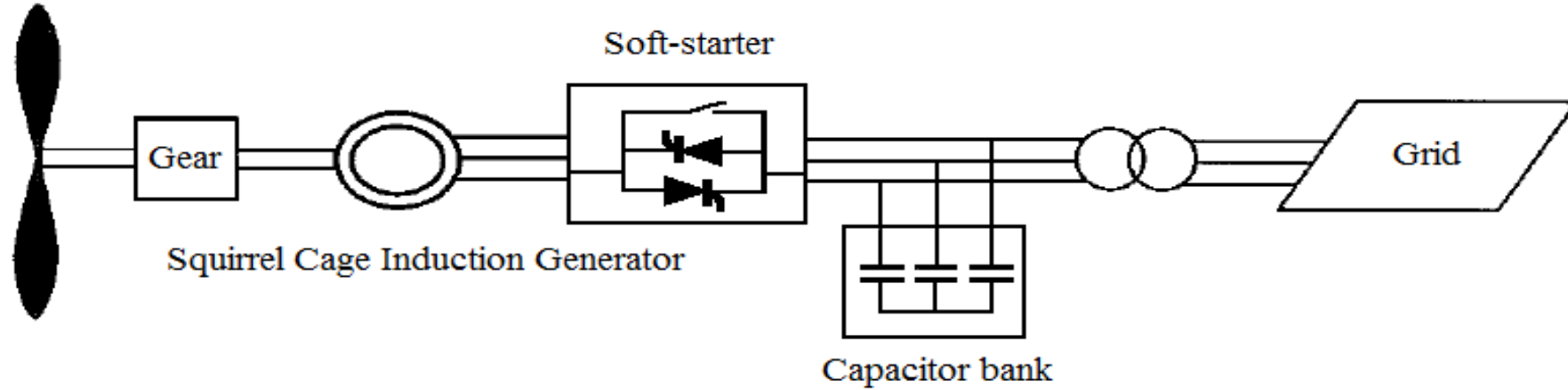
A wound-rotor, doubly-fed induction generator (DFIG)

Gearless Variable-Speed Synchronous Generator



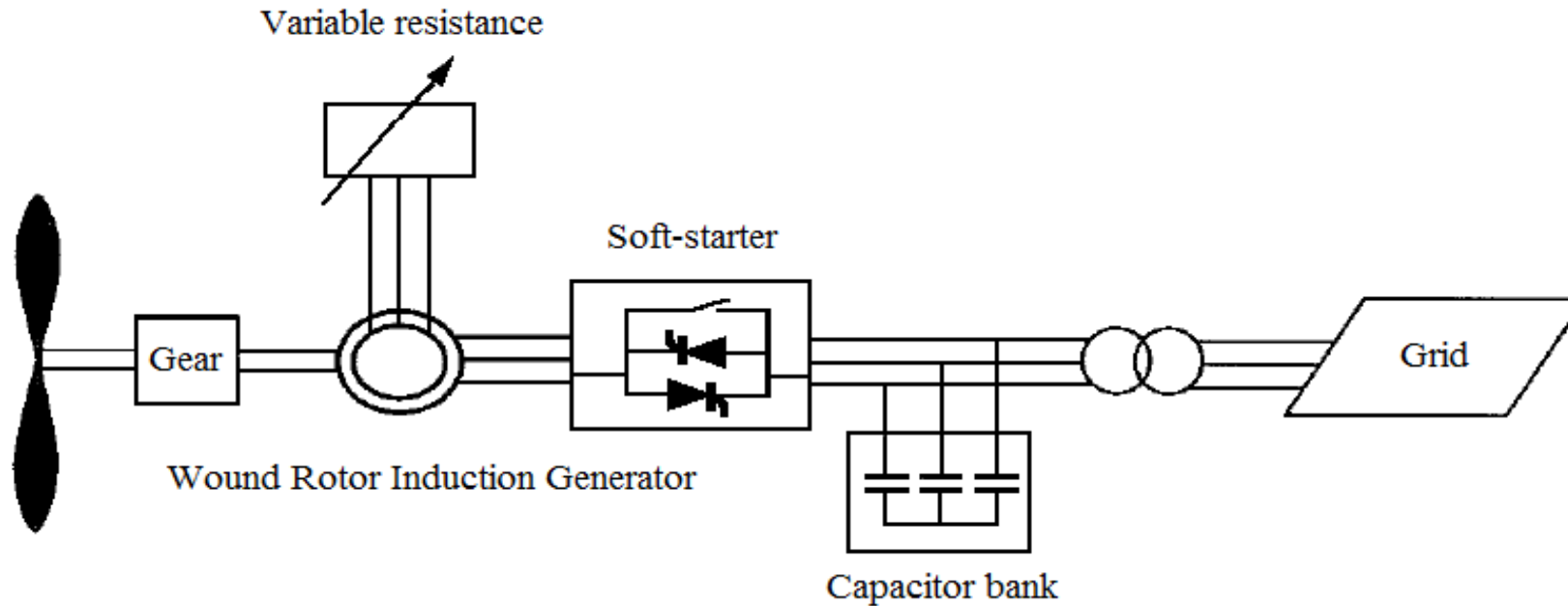
A gearless variable-speed synchronous generator

WECS Type A



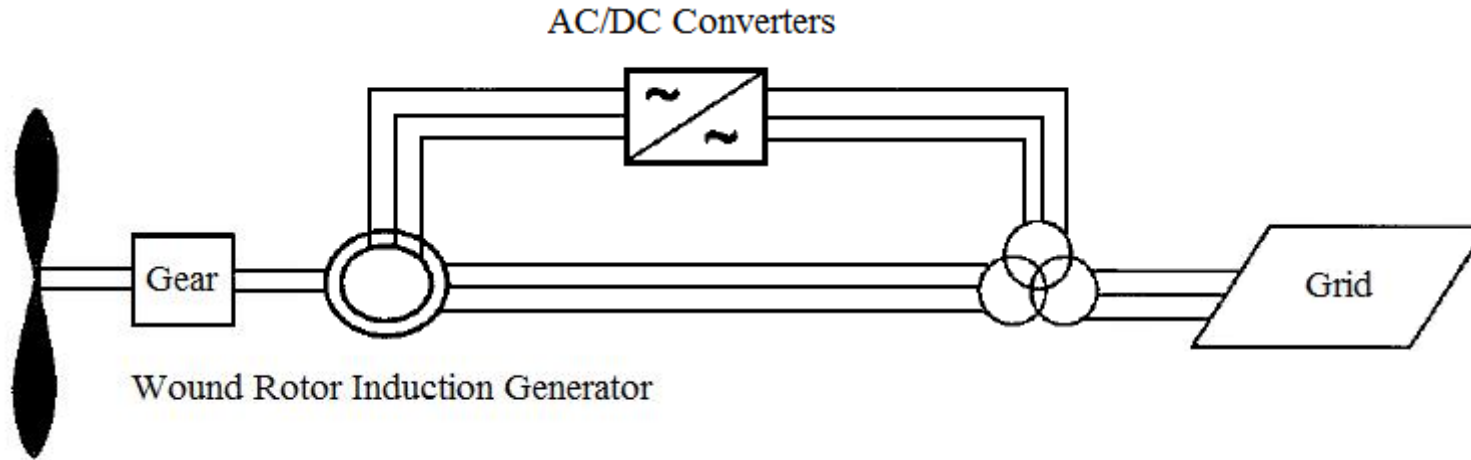
- Induction generator connected with a fixed-speed wind turbine
- Design requires 2 additional components for grid connection:
 - Soft-starter to decrease current transients during startup phase
 - Capacitor bank to compensate for reactive power.
- Capacitor bank enables the generator can work close to zero value generation and 0 reactive power consumption.
- However, this compensation approach does not provide flexible reactive power control.

WECS Type B



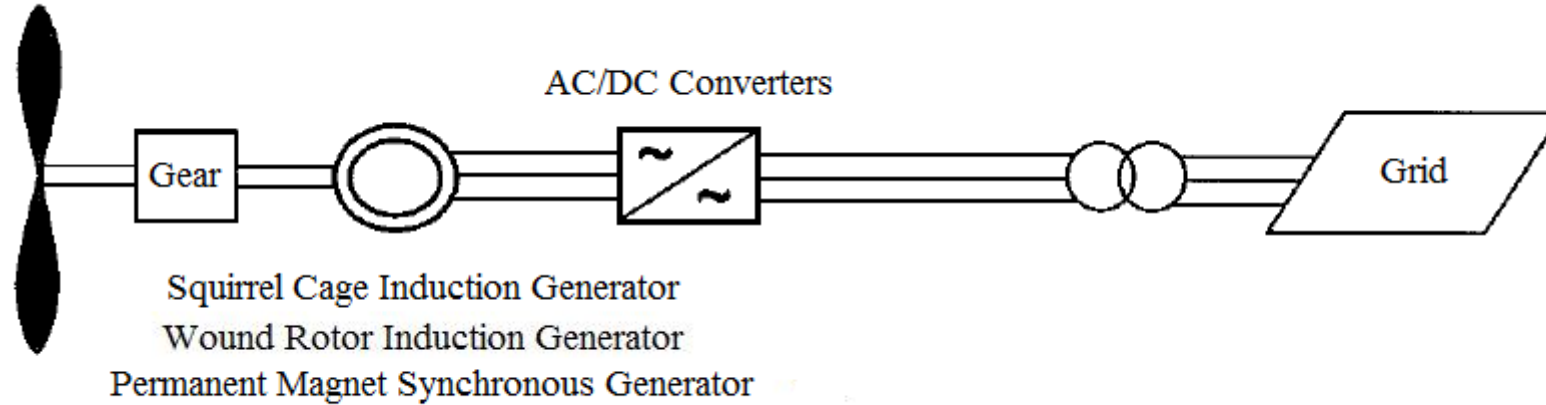
- Type B WECS generator is designed to work with limited variable speed wind turbine
- Variable resistor in the machine rotor, enables controlled-power output
- Capacitor bank and soft-starter are analogous to the type A design

WECS Type C



- WECS control enabled by two AC/DC converters w/ a connecting capacitor
- Wound rotor induction generator – known as a doubly fed induction generator (DFIG)
- “Doubly” as the rotor winding is not short-circuited (as in classical “singly-fed” induction machine); voltage is induced from the rotor-side converter
- 2 operating schemes: constant (1) reactive power or (2) voltage
- Most commonly installed WECS

WECS Type D



- Type D design includes full-scale frequency converter with different generator types.
- Most common – permanent magnet synchronous generator (PMSG).
- This design enables
 - full active / reactive power production control
 - high wind energy extraction value
- Full power control improves power and frequency stability and reduces the short circuit power.
- Most type D designs do not need a gearbox – a distinct advantage